
HPLTool *Ultimate*

User manual

A guide to the multi-purpose Windows service tool: power timer, network management, firewall, diagnostics and a fast file search engine.

HPL Tool Ultimate

described release: ver.3.09

Release date

2026-07-12

Platform

Windows 10 / 11 (desktop application)

Labelling rule: the interface is divided into four numbered sections (1 - Power, 2 - Network, 3 - Tools, 4 - File search). The **ADMIN** label marks a function that works correctly **only when the program is run as administrator**. The **PIN** label marks a function protected by the program's PIN code.

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1. About the program

HPL Tool Ultimate ver.3.09 [2026-07-12] is a multi-purpose Windows service tool that combines in one window:

- a **power timer** with voice announcements and a choice of action (shutdown, restart, hibernate, sleep, log off),
- **network** management: enabling/disabling the network interface, a continuous ping monitor, quick DNS switching (Google, Cloudflare, Quad9, DHCP presets),
- a **local network scanner** that identifies the device vendor from the MAC address (full IEEE OUI database, ~40,000 entries) **NEW in v3.09**,
- the **My data** window - a complete connection summary (public and local IP, MAC, gateway, DNS, ping tests),
- **blocking applications' internet access** through Windows Firewall rules,
- service tools: active TCP/UDP connections with process names, a system report (CPU, RAM, motherboard, disks + SMART), TEMP cleanup, DISKPART, Keep Awake,
- a **clipboard history** with pinnable entries,
- a very fast, index-based **file search engine**: by name, size, date, extension, combined criteria and file contents, with duplicate detection and batch renaming.

The interface is available in **Polish and English** (the PL / EN switch in the top-right corner takes effect immediately) and uses the dark "Windows 11 Dark Industrial" theme with a teal accent.

Requirements

- Windows 10 or 11.
- When run from the `.py` file: Python 3 with the **pyautogui**, **pyttsx3** and **Pillow** libraries. The EXE build ships with all dependencies.
- Internet access for the network functions (public IP, ping, DNS).
- Speakers/audio output for the voice announcements (Windows SAPI speech synthesizer).
- **Administrator rights** for some functions - the full list is in chapter 2.

2. Startup, PIN and administrator rights

Starting the program

1. Run the program (the `hpl_tool_search.py` file or the supplied executable).
2. The window opens at 1480x1000 (minimum 1100x700) with the four sections.
3. To use the functions marked **ADMIN**, close the program and start it again with **right-click** → **"Run as administrator"** (and confirm the UAC prompt).

Functions that require administrator rights

Important: the functions below execute system commands (`netsh`, `shutdown /fw`, `diskpart`) that Windows only allows with elevated rights. Without administrator mode the program shows an error such as "Error - run as administrator".

Function	Section	Without administrator
DISABLE NETWORK / ENABLE NETWORK ADMIN	2 - Network	the command is ignored - the interface state does not change
SET DNS ADMIN	2 - Network	the message "Error - run as administrator" appears
RESTART TO BIOS ADMIN	1 - Power	the <code>shutdown /r /fw</code> command ends with access denied
Firewall: BLOCK / UNBLOCK / DELETE rule ADMIN	3 - Tools	the message "Error creating/removing rule" appears
DISKPART ADMIN	3 - Tools	diskpart refuses to work in a non-elevated console
CLEAN TEMP - the <code>C:\Windows\Temp</code> part ADMIN	3 - Tools	only the user's TEMP and the Recycle Bin are cleaned; system files increase the error counter

All remaining functions (timer, ping, network scanner, My data, connections, system report, clipboard, the whole file search) work without administrator rights. Viewing the firewall rule list also needs no administrator - only deleting rules does.

PIN code

Sensitive operations are additionally protected by a PIN code **PIN**. The default PIN is `0000` - change it with the **CHANGE PIN** button in section 3.

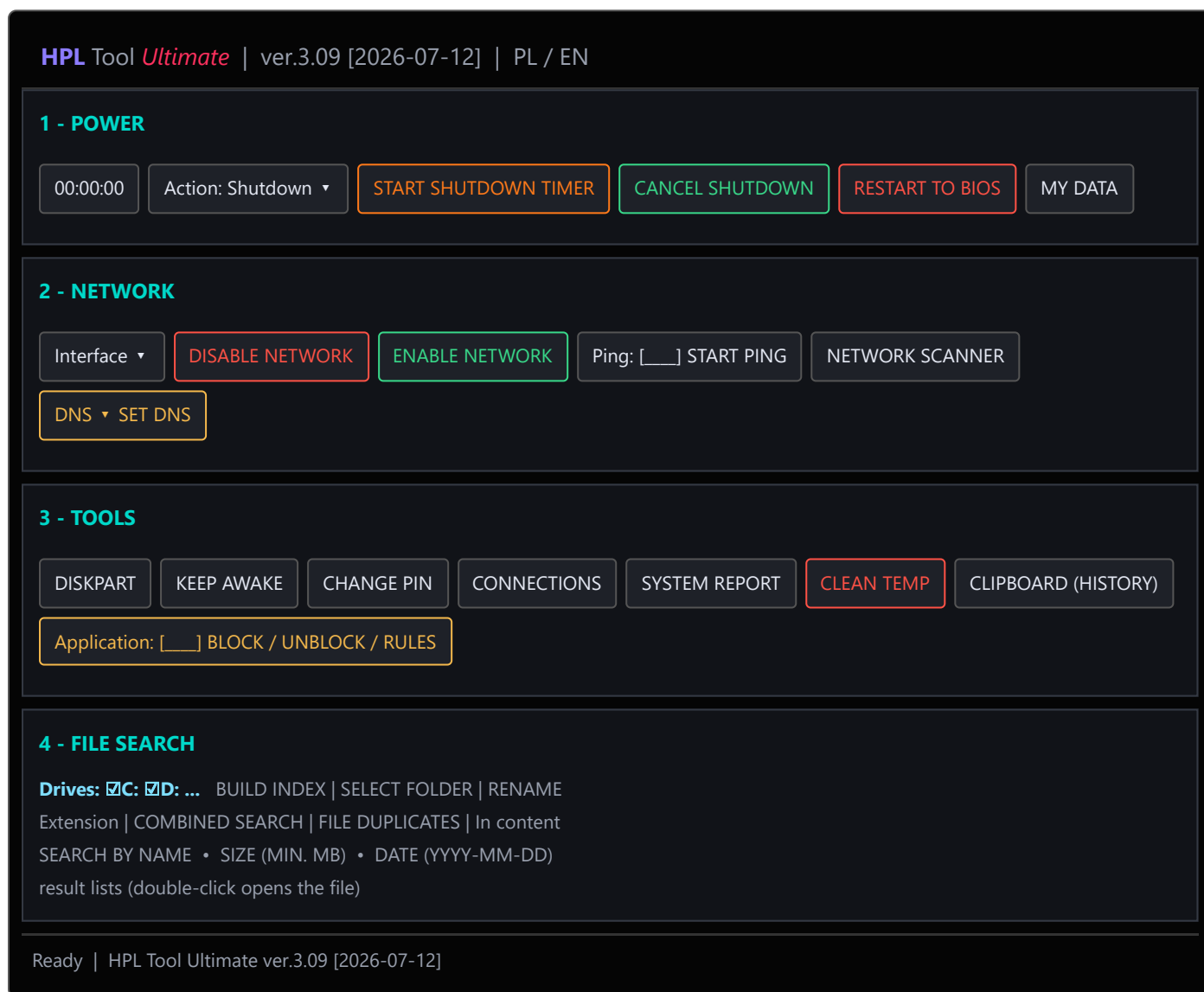
- The PIN protects: **closing the program**, cancelling the timer, restart to BIOS, enabling/disabling the network, DNS change, TEMP cleanup, firewall blocking and moving duplicates to the Recycle Bin.

- After **3 failed attempts** PIN entry is locked for **5 minutes**.
- The PIN is stored in the `pin.dat` file as a SHA-256 hash (it cannot be read back from the file).

Forgotten PIN close the program (e.g. via Task Manager), delete the `pin.dat` file from the program folder and start the program again - the PIN resets to `0000`.

3. Main window

The following drawing is a privacy-safe UI diagram. It contains no IP addresses or other user data.



- The **title bar** holds the HPL logo, the program name and the **PL / EN** language switch (instant, no restart needed).
- Sections 1-3 scroll inside the upper area of the window (mouse wheel); section 4 fills the rest of the window.
- **Closing the window (X) requires the PIN** - the program is also designed as a parental / service lock.

4. Section 1 - Power and timer

Power timer

1. From the **Action** list choose what happens when the countdown ends: *Shutdown, Restart, Hibernate, Sleep* or *Log off*.
2. Click **START SHUTDOWN TIMER** and enter the time in minutes (1-1440, i.e. up to 24 hours).
3. The large clock counts down in `HH:MM:SS` format and the button changes to "🕒 TIMER ACTIVE".
4. The program **speaks** voice announcements: 3 minutes before the end, 1 minute before the end and at the moment the action is executed (in Polish or English, matching the interface language).

CANCEL SHUTDOWN PIN stops the countdown and additionally runs `shutdown /a`, so it also aborts a shutdown scheduled outside the program. Starting the timer twice is blocked.

RESTART TO BIOS PIN ADMIN

Immediately restarts the computer straight into the UEFI/BIOS settings (`shutdown /r /fw /t 0`). Requires a UEFI motherboard and the program running as administrator.

Warning: the restart happens immediately, without asking to save open documents.

5. My data

The **MY DATA** button in section 1 collects a complete connection summary in the background (a few seconds) and opens a window with copyable rows:

Item	Description
Internet status	OK / Offline - a public-IP retrieval test.
Router ping / DNS ping / Internet ping	Three independent reachability tests: default gateway, first DNS server, the 8.8.8.8 server.
Public IP / Local IP / Subnet mask	The computer's addresses as seen from the internet and on the LAN.
Computer MAC / Computer name	Hardware identification on the network.
Default gateway / Router MAC / DNS servers	Router details and the DNS servers currently in use.

Every row has a **COPY** button; **COPY ALL** copies the whole report - convenient for support tickets.

6. Section 2 - Network: interface, ping, DNS

Enabling and disabling the network

PIN

ADMIN

Choose the network adapter from the **Interface** list (the program detects active interfaces), then use **DISABLE NETWORK** or **ENABLE NETWORK**. This is a hard, system-level disable of the adapter - the computer loses connectivity until it is enabled again.

Ping monitor

1. Enter an address (default `8.8.8.8`) and click **START PING**.
2. The program pings the address every second and shows the response time in colour: **green** below 50 ms, **teal** below 150 ms, **orange** above that, and a red **TIMEOUT** when there is no response.
3. **STOP PING** stops the monitor.

Quick DNS switching

PIN

ADMIN

Choose a preset from the **DNS** list and click **SET DNS**. Available presets:

- **Automatic (DHCP)** - back to the servers assigned by the router,
- **Google** (8.8.8.8 / 8.8.4.4), **Cloudflare** (1.1.1.1 / 1.0.0.1), **Quad9** (9.9.9.9 / 149.112.112.112).

The change applies to the adapter selected in the **Interface** list. The result ("OK: ..." or an error) appears next to the button.

7. Local network scanner

The **NETWORK SCANNER** button in section 2 finds all devices on the local network (a /24 subnet, up to 254 addresses). Scanning takes several seconds:

1. The program pings the whole subnet in parallel (batches of 64 addresses), filling the system ARP table.
2. It reads IP + MAC pairs from the ARP table (skipping broadcast/multicast addresses).
3. Device names are resolved in two stages: DNS first, then NetBIOS (`nbtstat`) for Windows computers without a DNS entry.
4. The device vendor is identified from the MAC prefix using the full **IEEE OUI** database (the `oui_db.txt` file, ~40,000 entries); without the file a built-in list of the most common vendors is used (Raspberry Pi, Espressif/smart home, TP-Link, Apple, Samsung, MikroTik, Ubiquiti and others). **NEW in v3.09**

Results go to an **IP | MAC | Name | Vendor** table. The column headers are clickable - ascending/descending sort with a direction arrow **NEW in v3.09**. Each row can be copied with **COPY**, and the whole list with **COPY ALL**.

Tip: your own computer always appears on the list (ARP does not show it; the program adds it itself). If the list looks suspiciously short, check Wi-Fi client isolation on the router and disable any VPN.

8. Section 3 - System tools

DISKPART **ADMIN**

Opens a console window with the `diskpart` disk and partition management tool. Diskpart requires administrator rights.

KEEP AWAKE

Prevents the computer from sleeping and the screensaver from starting: every 30 seconds it gently moves the mouse cursor to a random position. Clicking again turns the function off (the button changes colour while active).

CHANGE PIN **PIN**

After entering the current PIN, lets you set a new one.

CONNECTIONS

The **Active connections** window lists all TCP and UDP connections (`netstat`): protocol, local address, remote address, state and the **name of the process** that opened the connection. Handy for a quick check of "what is talking to the internet".

SYSTEM REPORT

A service report of the computer: operating system (with build number), CPU, RAM amount, motherboard model and serial number, and the list of physical disks with size and **SMART / health status** (OK, Healthy, Warning...).

CLEAN TEMP

PIN

ADMIN — partially

After confirmation deletes temporary files from `%TEMP%` and `C:\Windows\Temp`, and empties the **Recycle Bin**.

The result ("Deleted: N files | Errors: M") appears next to the button. Cleaning `C:\Windows\Temp` fully works only in administrator mode; files currently in use by running programs are skipped and increase the error counter - this is normal.

9. Clipboard history

The program continuously monitors the text clipboard (about every 0.8 s) and saves the history to the `clipboard.json` file. The **CLIPBOARD (HISTORY)** button opens the manager:

- the **30 most recent entries** are kept, plus **pinned** 📌 entries (no limit) - pinned entries never drop out of the history,
- 📌 pins/unpins an entry (pinned entries always stay at the top of the list),
- **COPY** puts the entry back into the clipboard and closes the window,
- ✕ deletes a single entry, and **CLEAR HISTORY (KEEP PINNED)** removes everything except pinned entries,
- copying a text that is already in the history moves it to the top and keeps its pin.

10. Blocking applications in the firewall

The **Application** row in section 3 blocks a chosen program's internet access using Windows Defender Firewall rules.

Blocking PIN ADMIN

- 1. Click **BROWSE** and point to the application's `.exe` file (or type the full path manually).
- 2. Click **BLOCK**. The program creates two firewall rules - for outbound and inbound traffic - named `HPL_BLOCK_<name>_OUT` and `HPL_BLOCK_<name>_IN`.
- 3. **UNBLOCK** removes both rules for the given application.

Rule overview

RULES opens a list of all `HPL_BLOCK_*` rules present in the firewall. Viewing alone does not require administrator; the **DELETE** button next to a rule PIN ADMIN removes it individually.

Note: firewall rules stay in the system after the program is closed - a blocked application has no internet until the rules are removed (with UNBLOCK, from the RULES list, or manually in Windows Firewall).

11. Section 4 - File search

Two operating modes

Mode	When active	Characteristics
Index	when no folder is selected (the label reads "All drives")	Instant results from a previously built index of all ticked drives.
Live	after clicking SELECT FOLDER	Searches the chosen folder directly on disk - always up to date, slower. The X button returns to index mode.

Building the index

- 1. Tick the drives to index (the checkboxes next to "Drives:"; network drives are detected too).
- 2. Click **BUILD INDEX**. The program scans the drives in the background and saves a catalogue of all files (path, size, date) to the `index.json` file.
- 3. When finished, a message with the file count appears together with the spoken "Indexing complete".

Tip: the index is a snapshot - new or deleted files are not visible in it until you rebuild it. Refresh the index after larger changes on the disks.

Basic searches

- **SEARCH NAME** - a phrase in the file name; supports wildcard masks, e.g. `*.iso` or `report*2026*`. Results in the left list.
- **FILTER SIZE** - files larger than the given threshold in MB (default 500). Results with a `[MB/GB]` size in the right list.
- **SEARCH BY DATE** - files modified on the given day *or newer* (YYYY-MM-DD format).
- **SEARCH EXTENSION** - all files with the given extension (e.g. `mkv`).

After each search finishes, the program says "Search complete".

12. Combined search, content search and result menu

COMBINED SEARCH

Combines **all filled-in** criteria at once (AND): the name phrase, extension, minimum size and date. Example:

+ min. 4000 MB + date 2026-01-01 finds large movies added this year. An empty field = the criterion is skipped.

SEARCH IN CONTENT

1. First **SELECT FOLDER** - content search works only in the selected folder.
2. Enter a phrase in the "In content:" field and click **SEARCH IN CONTENT**.
3. The program searches text files (txt, log, ini, cfg, csv, xml, json, md, py, bat, ps1, html, css, js, sql, yaml, srt, nfo, reg and others), skipping files larger than 20 MB; the result limit is 1000 hits.
4. Results in the left list are grouped by file, with a line number and a content preview; double-clicking opens the file.

Working with results

- **Double-click** opens the file in its default program.
- **Right-click** on a result opens a menu: open the file, **open the containing folder** (with the file selected in Explorer) and **file properties**.

13. File duplicates

FILE DUPLICATES analyses the index and groups files with an **identical name and identical size**. Requires a built index.

1. The window opens on the right half of the screen; groups are sorted from the largest files (at most 200 groups shown).
2. Each group shows the name, size and number of copies; **SELECT COPIES** ticks all files in the group *except the first one* (the original stays).
3. The counter at the bottom sums the selected files and their size.
4. **DELETE TO RECYCLE BIN** moves the selected files to the system Recycle Bin (they can be restored). The program verifies what actually disappeared from disk, updates the index and refreshes the view.

Warning: the name+size criterion does not compare file contents. Before deleting, make sure the selected entries really are copies.

14. Batch rename and program files

RENAME

- 1. First **SELECT FOLDER** - renaming works on the files of the selected folder (without subfolders).
- 2. Set the operations: **Replace** one text with another, add a **Prefix** and/or **Suffix**, optionally **Number (001, 002...)**.
- 3. The live **OLD** → **NEW** preview shows the outcome for every file; extensions remain untouched.
- 4. Confirm with the rename button - the program reports the number of renamed files.

Program files

File	Purpose
pin.dat	SHA-256 hash of the PIN code. Deleting the file restores the 0000 PIN.
index.json	The search engine's file index (paths, sizes, dates). Updated when duplicates are deleted.
clipboard.json	Clipboard history (atomic saving - the file survives crashes undamaged).
oui_db.txt	The full IEEE OUI vendor database for the network scanner (optional; without it the built-in list is used).
icon.ico, logo3_HPLInk.jpg	The window icon and the logo in section 1 (optional).

15. Troubleshooting

"Error - run as administrator" (DNS, firewall, network)

The function needs elevated rights. Close the program (PIN) and start it again with right-click → **"Run as administrator"**. The full list of admin-only functions is in chapter 2.

DISABLE NETWORK does nothing

The `netsh interface set interface` command is silently rejected by the system without administrator rights. Run the program as administrator. Also check that the correct interface is selected in the list.

Forgotten PIN / PIN locked

After 3 failed attempts wait 5 minutes. If the PIN is forgotten - delete the `pin.dat` file from the program folder (close the program first, e.g. with Task Manager); the PIN resets to `0000`.

The search shows "Index not found"

In "All drives" mode the search uses the index - click **BUILD INDEX** first. Alternatively pick a folder with **SELECT FOLDER** to search live without an index.

Search results are out of date

The index is a snapshot from the moment it was built. Rebuild the index after larger changes on the drives.

The network scanner finds few devices

Temporarily disable any VPN, check client isolation (AP isolation) on the router and scan again - the ARP table fills up with each pass. Sleeping devices may not answer pings.

No voice announcements

Check the audio output and system volume. The synthesizer uses Windows (SAPI) voices - verify in Windows Settings that speech voices are installed.

CLEAN TEMP reports errors

Files currently used by running programs cannot be deleted - the "Errors" counter is normal then. Full cleaning of `C:\Windows\Temp` requires administrator mode.

RESTART TO BIOS returns an error

The function requires running as administrator and a UEFI motherboard. On older (legacy BIOS) computers the `shutdown /r /fw` command is not supported.

A blocked application still has internet access

Check in the **RULES** window whether the `HPL_BLOCK_*` rules exist (if not, the block was not created - most likely due to missing administrator rights). Also make sure the correct `.exe` path was blocked - some programs have several executables.